

ENGR 213 Final Exam Review – Fall 2020

Premium

Course: Applied Ordinary Differential Equations (ENGR 213) 999+ documents

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FinalExam

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Problem 1. Using an appropriate substitution transform the differential equation below such that the separation of variables method can be used to solve the initial value problem

$$y' = (y - 7x)^2 - 18, \quad y(0) = 1$$

Problem 2. Given the following differential equation

$$\left(-\cos(xy^2) + \frac{x^2}{y} - xy^2 \sin(xy^2) \right) dx - \left(2yx^2 \sin(xy^2) + \frac{x^3}{y^2} \right) dy = 0$$

and the initial condition $y(\pi) = 1$.

- (a) Find the integrating factor which makes the above equation exact.
- (b) Use the exact equation method to find the differential equation's solution in implicit form.
- (c) Solve the initial value problem defined in this problem.

Problem 3. It is believed that life appeared on the Earth when two sets of proteins merged together to form the first life form. Let us call this life form C and the two proteins A and B . To form life, 2 grams of A and 3 grams of B are needed to form 5 grams of C . In a pond 50 gram of C is formed after 60 minutes. Assume that the rate of reaction is proportional to the product of remaining amounts of A and B . If 100 gram of A and 80 gram of B were initially present in the pond, find

- (a) the amount of C as a function of time.
- (b) How much life form (in grams) is present after 1 hour and 15 minutes.

Problem 4. Solve the initial value defined by the following differential equation

$$t^3 \frac{dy}{dt} + y^2 = t^2 y,$$

subject to the initial condition $y(2) = 4$.